

Priyadarshini College of Engineering, Nagpur
Sessional Examination (2023-24) Even Semester
B. Tech. Eight Semester (Computer Technology) (C.B.C.S)
Social Network (Elective-VI)

P. Pages: 2

PCE/KS/24/BTCT801T.1

Time: Three Hours

Max. Marks: 70

Notes:

1. All questions carry marks as indicated.
2. Solve Question 1 or Question 2.
3. Solve Question 3 or Question 4.
4. Solve Question 5 or Question 6.
5. Solve Question 7 or Question 8.
6. Solve Question 9 or Question 10.
7. Due credit will be given to neatness and adequate dimensions.
8. Assume suitable data wherever necessary.
9. Illustrate your answers whenever necessary with the help of neat sketches.

Q. No.	Question	CO	B T	Marks
1.	a) Discuss Social Network in detail.	CO1	2	7M
	b) Describe Networkx in detail.	CO1	2	7M
	OR			
2.	a) Describe following Social Network dataset formats: 1. CSV 2. GML 3. Pajek 4. GraphML 5. GEFX.	CO1	2	7M
	b) Enlist the challenges in social network.	CO1	2	7M
3.	a) Discuss Fatman Evolutionary model for the base code for adding a people.	CO2	2	7M
	b) Describe Granovetter's strength of weak ties.	CO2	2	4M
	c) Elaborate triadic closure.	CO2	2	3M
	OR			
4.	a) Illustrate community detection using Girvan Newman algorithm.	CO2	2	7M
	b) Explain validation of Granovetter's experiment using cell phone data.	CO2	2	7M
5.	a) Discuss fact equal coin distribution.	CO3	2	7M
	b) Summarize impact of communities on diffusion of Network.	CO3	2	7M
	OR			
6.	a) Explain how to find google page ranking using web graph.	CO3	2	7M
	b) Describe web graph in social network.	CO3	2	7M

Q. No.		Question	CO	B T	Marks
7.	a)	Define Power law and explain richer-get richer phenomenon.	CO4	2	7M
	b)	Analyze the connection between power law and the spread of epidemics.	CO4	3	7M
		OR			
8.	a)	Discuss implications of power laws in social network by using suitable example.	CO4	2	7M
	b)	Compare SIR and SIS spreading models.	CO4	2	7M
9.	a)	Describe small world phenomenon in social network.	CO5	2	7M
	b)	Summarize the concept of going viral on the web.	CO5	2	7M
		OR			
10.	a)	Discuss the key concept and application of network analysis in real world scenario.	CO5	2	7M
	b)	Describe Decentralized search in Small World Effect.	CO5	2	7M
